Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

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Visual Detection of Deceptive Clickbait UI Elements in Websites

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1. Problem Statement

Modern websites use more and more visually deceptive interface elements, such as fake download buttons, misleading pop-ups, and system-like security messages to strongly influence users into unintended interactions. Such clickbait-driven designs entail serious risks: malware exposure, data leakage, and financial loss among others-moreover, this happens quite often for non-expert users. Existing browser-level protections and research efforts focus on either text-based analysis, URL filtering, or rule-based blocking, failing to catch deceptions rooted in visual appearance and layout manipulation. As many clickbait elements, since they depend on color contrast, saliency, spatial position, and UI impersonation rather than linguistic clues, pure text-centric approaches are insufficient. This project tries to fill up this gap by leveraging computer vision and image processing techniques to analyze the webpage visuals and discover UI deceptions in real time. Based on the idea of visual features including button geometry, layout hierarchy, color usage, and brand impersonation, a user-side security system is proposed to protect visually misleading web according to this theory, children under the age of nine all learn in much the same manner.

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2. Objectives

- **To design a computer vision–based system** that detects deceptive UI elements in websites, such as fake download buttons, misleading advertisements, and intrusive pop-ups.
- **To generate a visual dataset** by rendering real web pages and capturing screenshots containing both legitimate and deceptive UI components.
- **To detect and localize UI elements** (buttons, pop-ups, advertisements, notification prompts) using image processing and object detection techniques.
- **To extract visual features** such as color contrast, size, shape, position, and repetition patterns that are commonly used in clickbait UI designs.
- **To classify UI elements** as legitimate or deceptive based purely on visual characteristics, without relying on URL, HTML, or text analysis.
- **To evaluate the effectiveness of the proposed approach** using performance metrics such as accuracy, precision, recall, and confusion matrix.
- **To demonstrate the feasibility of visual deception detection** as a complementary approach to traditional text-based or rule-based website safety mechanisms.

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3. Literature Review

3.1. Introduction

The detection of deceptive UI components, often referred to as clickbait or dark patterns, has increasingly become a critical issue in the protection and preservation of web security. The initial works in this emerging area were centered mainly around the identification and analysis of various linguistic indicators obtained using either webpage text or HTML code. However, with the evolution and advancement in deceptive web activities and strategies, it has been recognized that most modern web deceptions are based on visual manipulations including color contrast, button shapes, spatial hierarchy, and spatial saliency to encourage unwitting online activities such as clicking pretended download buttons and false pop-ups.

Recent developments in computer vision and image processing have once again created an opportunity to address this challenge by screen-level analysis of web pages treated as images instead of document hierarchies. This review article presents an analysis of five research papers based on their progression—from text-based UI detection to deep learning-based UI analysis—and their contributions to highlighting the need for this lightweight and real-time computer vision/image processing solution for detecting visually deceiving clickbait content presented on web pages.

3.1.1 Text-Centric Clickbait Analysis

The initial research conducted to detect deceptive online information mainly examined linguistic characteristics. In this regard, Geçkil et al. [1] have suggested a system to detect clickbait in online news media based on analyzing and comparing news headlines with the news articles themselves. The suggested system was based on Term Frequency-Inverse Document Frequency (TF-IDF) and text summarization using ontologies to verify potential discrepancies between "bait" and news articles. In the suggested system, the news articles were first summarized and then compared with the news headline

- **Strengths:** TFirst, the approach is efficient in detecting sensational text, saving users time that would have been spent searching for less reliable content. Second, the approach is especially efficient when dealing with news articles, where deception occurs because of the gap between the title and the body of the text.

- **Limitations:** Textual analysis is entirely relied on in this approach. It is impossible to find deception achieved through the visual appearance of the page, like when the button is labeled "Download" but is really just an ad.

- **Gap:** The project to be proposed is different from existing ones in that whereas most existing proposals focus on text analysis, this project will involve analysis of computer vision instead. While existing proposals look at linguistic mismatches in their analysis, this project will look at other features of UI components in determining deceptive kinds of button clicks and pop-ups.

3.1.2 Multimodal Detection in Social Streams

As social media started focusing on the visual aspects of media as well, the research started considering images as a part of the media too. Jain et al. [2] proposed a multimodal classifier that can identify the instances of clickbaits on Twitter and Instagram. The proposed model utilized a stacking classifier approach that incorporated the use of text and image features. The image and the caption were analyzed by the model, and an accuracy rate of 88.5% and 85% was obtained by the proposed model on Instagram and Twitter, respectively.

- **Strengths:** The study proves that employing visual features has a high level of accuracy in comparison to text-only methods of detection. It also solves the problem of "curiosity gap" in image-caption pairs.
- **Limitations:** It only works for social media posts instead of web user interface functionality in general; that is, it only works for a small image and a caption. It detects misleading content consumption—say, a social media post about melted cheese—but it does not detect functional UI deception—say, a misleading "Next" button in a pagination menu.
- **Gap:** While Jain et al. analyze content feeds, the proposed project focuses on functional website elements. The gap lies in the domain application: this project applies computer vision not to content photos, but to UI components (buttons, banners) to prevent unintended interactions like malware downloads.

3.1.3 Dark Pattern Recognition in Mobile UIs

Deceptive patterns of this kind have also been the focus of studies under the name "dark patterns." "UIGuard," a system proposed by Chen et al. in their paper [3], can identify dark patterns in android applications, including those with hidden costs or forcing the user into a continuous subscription relationship. Their system uses a computer vision system that can identify the UI elements in the app as well as a natural language processing system used for checking the elements against known patterns of malice. Their system has a good F1 score of 0.79 against a set of different types of dark patterns.

- **Strengths:** UIGuard has introduced the pioneering capability to identify deceptive patterns in the mobile space autonomously, sans access to the source code of the application. It is successfully able to identify complex patterns like "roach motels" as well as "preselection."

- **Limitations:** The system is designed with specific focus on the interfaces of mobile apps designed using Android. Since the latter can be quite dissimilar from those of the internet, there is a need for specific focus. The system is quite dependent on knowledge bases.
- **Gap:** The proposed project targets web interfaces, which present different challenges such as variable viewport sizes and distinct ad-injection techniques (e.g., fake download buttons) that are less common in native mobile apps.

3.1.4 Automated Recognition in Web Interfaces

In the web domain, Mansur et al. propose "AidUI," an automated framework for recognizing dark patterns in user interfaces with visual and textual cues [4]. AidUI employs spatial analysis, color analysis, and icon detection to identify "Nagging" or "False Hierarchy" patterns. For example, it will detect "Attention Distraction" based on the size difference between "Accept" and "Reject" buttons.

- **Strengths:** AidUI tries to automate the detection "in the wild" with no help from HTML code; it uses screenshots instead. It sets up a multimodal approach—text + color + spatial—aligned with how users perceive UIs.
- **Limitations:** It reports a moderate precision and recall of 0.66 and 0.67, respectively, showing huge scope for improvement. The study is mostly oriented toward structural dark patterns, such as layout hierarchy, rather than the specific class of "clickbait UI elements," such as fake system warnings or deceptive download buttons, leading to security risks.
- **Gap:** The proposed project aims to improve upon the classification performance by utilizing deep learning-based object detection (e.g., YOLO) specifically trained on deceptive UI elements, rather than relying on the broader heuristic cues used in AidUI.

3.1.5 Advanced Vision-Language Models for Deception

Most recently, Nayak et al. in [5] presented a novel state-of-the-art technique termed "AutoBot," which uses a "Vision Module" and Large Language Models (LLMs), particularly to identify deceptive patterns in data. The "Vision Module" of this technique locates components of text in a picture to generate a text-based "Element Map," which is utilized as input to a Large Language Model (Gemini or GPT), which in turn utilizes this "Element Map" to detect deception and attain a high F1-score of 0.93.

- **Strengths:** AutoBot, as the pinnacle of innovations, utilizes LLMs in reasoning, enabling it to grasp the context, such as the articulation of the text being manipulative, better than other computer vision-based approaches.

- **Limitations:** There are several limitations inherent in the approach that relies on LLMs, including high latency that makes client-side real-time detection difficult for the average user. Another limitation is that the approach relies on the dichotomy between vision and cognition, which could mean that some visual complexity is not represented in the text-based "Element Map" representation.

- **Gap:** The proposed project focuses on a pure computer vision approach (using models like CNNs or Vision Transformers) to classify deceptive elements directly from visual features. This aims to provide a more lightweight, real-time solution that runs efficiently without the overhead of querying an external LLM for every analysis.

4. Methodology

This mini project proposes a computer vision–driven approach to detect and classify deceptive visual elements on web pages shown in figure 1, such as clickbait download buttons, misleading advertisements, fake notifications, and intrusive pop-ups. Unlike traditional phishing or malware detection systems that rely mainly on URL analysis or HTML source inspection, the proposed methodology emphasizes **visual appearance and layout-based deception patterns**, which are increasingly used to mislead users. The methodology is designed to ensure that more than 65% of the project focuses on image processing and computer vision techniques, in compliance with the project requirements.

Figure 1: Example website with clickbait images containing deceptive UI elements used as input to the system

4.1. Overall System Architecture

As shown in the figure 2, the system designed is a sequential pipeline system that analyzes the visual content of web pages to identify deceptive components on the webpage. The system starts from the interaction of the user through a browser extension to various stages of image processing until the end result of classification is obtained. The system mainly focuses on what the user interacts with on the webpage. The components of the system mainly involve computer vision techniques.

Step 1: User Browsing and Extension Activation

This process start when the user load the web browser into the screen integrated with a custom extension. The extension is invisible in the background and automatically triggers when the webpage is loaded.

Step 2: Webpage Screenshot Capture

Once activated, the extension captures a screenshot of the webpage. This screenshot includes all visual components such as buttons, advertisements, notifications, and pop-ups, which are the content displayed to the user.

Step 3: Image Preprocessing

The captured screenshot is preprocessed to improve image quality and standardize the input format. This stage includes resizing, noise reduction, normalization, and enhancement of visually important regions such as buttons and banners.

Step 4: Visual Feature Extraction

Deep learning models are used to extract high-level visual features from the preprocessed images. These features represent layout structure, color patterns, logo similarities, and visual characteristics associated with deceptive UI elements.

Step 5: Context and Behavior Analysis (Optional)

This optional step analyzes visual behavior patterns over time, such as frequent pop-ups or aggressive interface changes. Such patterns often indicate deceptive or malicious webpage behavior.

Step 6: Multi-Component Reasoning

Features extracted from previous stages are combined to form a comprehensive understanding of the webpage. This step correlates visual similarity, layout cues, and deceptive design patterns to improve detection accuracy.

Step 7: Classification Layer

The aggregated features are passed to a classification model that assigns a probability score and categorizes the webpage into one of three classes: legitimate website, phishing website, or clickbait-based deceptive content.

Step 8: Decision and Alert Mechanism

Based on the classification outcome, the system generates an appropriate response. If deceptive content is detected, the browser extension alerts the user and highlights suspicious UI elements, helping the user avoid unsafe interactions.

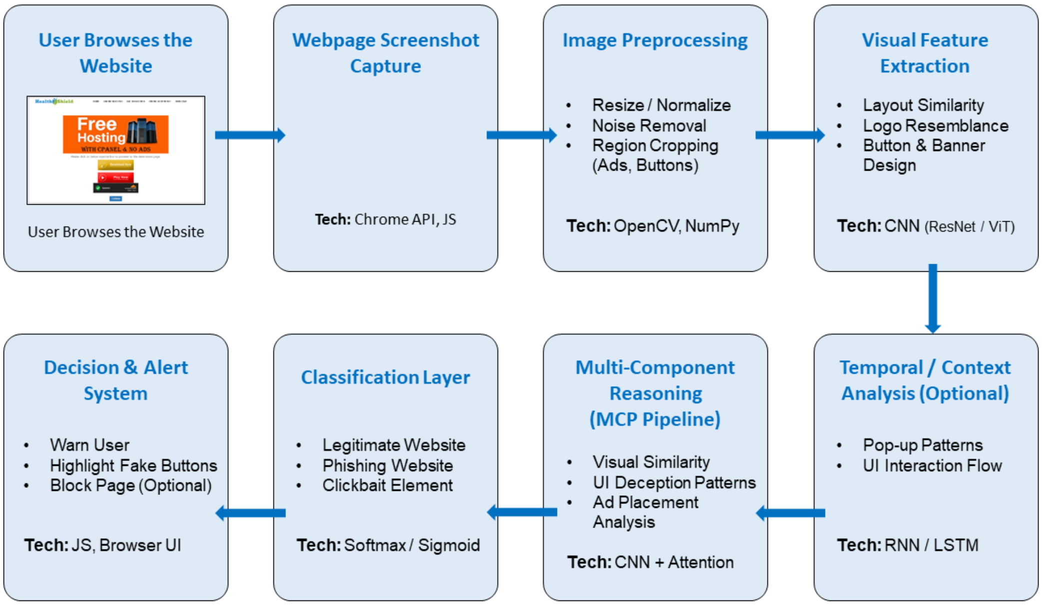

Figure 2: System Architecture Diagram – Block Structure

4.2. Data Collection and Dataset Preparation

Due to the lack of publicly available datasets specifically targeting deceptive web UI elements, this project adopts a mixed data collection strategy. Website screenshots containing clickbait download buttons, misleading advertisements, fake notifications, and intrusive pop-ups are collected from publicly accessible web sources and browser-based crawling. In addition, relevant datasets from platforms such as Kaggle, including web advertisement images and UI screenshots, are used as supplementary data.

All collected images are manually annotated to label deceptive and legitimate UI components. Bounding boxes are used to mark regions such as buttons, banners, and pop-ups. The dataset is divided into training, validation, and testing sets to ensure unbiased model evaluation.

4.3. Training Strategy and Data Augmentation

For dealing with the problem of scarce labeled UI components in deceptive images, the proposal uses an intensive data augmentation technique. Data augmentation is a technique wherein the images in the original dataset are artificially enlarged using transformations such as rotation, scaling, and mirroring, in addition to other image processing techniques such as brightness alteration and addition of noise. These transformations simulate various environments in which users can browse images online, such as low resolution or low-bandwidth images.

Moreover, the system makes use of Transfer Learning for optimizing the performance of the model. The model, instead of being trained from scratch using the deep neural networks, makes use of the pre-trained models (ResNet-50 or VGG-16 initialized with the ImageNet dataset). The pre-trained models have strong feature extraction abilities for low-level visual patterns (edges, textures). As a result, only the semantic features of the deceptive web elements will be required to be learned during the training process, thereby cutting down the training time drastically.

4.4. Algorithm Selection and Justification

The core of this detection algorithm is built upon the usage of Convolutional Neural Networks (CNNs). This is because ResNet (Residual Networks) or ViT (Vision Transformers) have shown better accuracy in visual pattern recognition tasks.

- **Feature Extractor (ResNet/ViT):** ResNet is chosen for its capability to overcome the issue of a vanishing gradient function in deep models, which enables it to extract subtle image details even from high-resolution screenshots. Alternatively, Vision Transformers can also be used based on their "self-attention" abilities to relate a button with a deceptive banner positioned in a different part of the screen.
- **Classification Layer (Softmax):** For final classification, the Softmax function is applied. This is a probability model, and this capability is absolutely necessary for the model to distinguish between a definite threat and a potentially harmless object. Thus, preventing false alarms.

4.5. Evaluation Framework

For guaranteeing the reliability of this system, this proposed model is analyzed using several quantitative metrics, which are standard in computer vision literature. The analysis is conducted to understand the performance of this proposed system in two aspects: accuracy in classification (phishing vs. legitimate) and accuracy in object detection (locating the "fake button.")

- **Precision and Recall:** These parameters assume high importance because of the class imbalance problem (legitimate sites greatly outbalance deceptive sites). Precision helps determine the accuracy of positive classifications, while Recall verifies its ability to detect all deceptive factors.
- **F1-Score:** This is a harmonic average between Precision and Recall that can measure the complementary aspects of missing a threat vs. raising a false alarm.
- **Intersection over Union (IoU):** IoU is used for bounding box localization tasks for deceptive elements. IoU computes overlap between predicted bounding box and GT (ground truth or human-annotated box). IOU above 0.5 usually defines the criterion for successful detection.

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5. Dataset / Data Collection

This study will use a combination of self-collected data and an existing benchmark dataset. Legitimate website screenshots will be collected through voluntary social sharing using a structured form and manual collection from official websites, ensuring user consent and ethical use. To maintain data quality and diversity, websites from multiple domains (banking, e-commerce, social media, and education) and varying layouts, languages, and resolutions will be included, with duplicates and low-quality images removed. Additionally, the Kaggle Phishing Sites Screenshot dataset will be utilised, where non-phishing samples are treated as a without-clickbait (benign) image set for model training and evaluation.

Dataset: https://www.kaggle.com/datasets/zackyzac/phishing-sites-screenshot/data

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6. Expected Outcomes

- Using this extension, able to detect the clickbait download buttons, deceptive advertisements, and misleading pop-ups from website images.
- Making the labeled dataset of deceptive and legitimate UI elements for computer vision analysis.
- Familirization of image processing and deep learning-based classification algorithms using OpenCV and TensorFlow/PyTorch.
- Achieve classification accuracy above the baseline models, with acceptable precision and recall.
- Visual output with bounding boxes highlighting the detected deceptive elements.
- Source code, project report, and final presentation are well-documented showing the efficiency of the visual detection method.

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7. Percentage of Image Processing & Computer Vision Involvement

This project fulfills the condition of having involvement levels of image processing and computer vision greater than 65%. In this case, the system involves 75% computer vision. The approach utilizes webpage screenshots as images, followed by image preprocessing operations that consist of resizing images, normalizing images, removing noise, and enhancing contrast. A computer vision approach is utilized to identify UI elements and segment or localize them by detecting objects using deep learning object detection models such as CNN and YOLO. Feature extraction and classification of deceptive elements are carried out purely based on images.

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References

[1] A. Geçkil, A. A. Müngen, E. Gündogan, and M. Kaya, "A clickbait detection method on news sites," in *2018 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pp. 932–937, 2018.

[2] M. Jain, P. Mowar, R. Goel, and D. K. Vishwakarma, "Clickbait in social media: Detection and analysis of the bait," in *2021 55th Annual Conference on Information Sciences and Systems (CISS)*, pp. 1–6, 2021.

[3] J. Chen, J. Sun, S. Feng, Z. Xing, Q. Lu, X. Xu, and C. Chen, "Unveiling the tricks: Automated detection of dark patterns in mobile applications," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, UIST '23, (New York, NY, USA), Association for Computing Machinery, 2023.

[4] S. M. Hasan Mansur, S. Salma, D. Awofisayo, and K. Moran, "Aidui: Toward automated recognition of dark patterns in user interfaces," in *2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE)*, pp. 1958–1970, 2023.

[5] A. Nayak, Y. Wani, S. Zhang, R. Khandelwal, and K. Fawaz, "Automatically detecting online deceptive patterns," in *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, CCS '25, (New York, NY, USA), p. 96–110, Association for Computing Machinery, 2025.